

## 6 Density Estimation

### 6.1 Histogram

- The histogram is our first proposal as an estimator for the density function.
- Area under the histogram is one.
- Area of each bar is the empirical estimate of the probability of the random variable being between the endpoints of that bar.
- In that sense it is an empirical estimate.
- But it is far from a smooth curve.
- The number and location of the bars are subjective. So it is not really a well-defined function of the data as an estimator should be.
- Choose bin width by minimizing MSE.

### 6.2 MSE of histogram

Each bin height is like

$$\hat{f}(t) = \frac{\hat{F}(t+\delta) - \hat{F}(t)}{\delta}$$

The distribution is  $\frac{1}{\delta t}$  times a binomial  $(n, p_t)$

$$p_t = \int_t^{t+\delta} f(x)dx$$

$$Var(\hat{f}(t)) = \frac{1}{\delta^2} \left( \frac{p_t(1-p_t)}{n} \right)$$

**Make  $\delta$  big to reduce variance**

For small  $\delta$ ,

$$E(\hat{f}(t)) = \frac{1}{\delta} p_t = \frac{1}{\delta} \int_t^{t+\delta} f(x)dx$$

By Taylor expansion,

$$F(t+\delta) = F(t) + \delta f(t) + \frac{\delta^2}{2} f'(t) + \dots$$

Thus

$$\int_t^{t+\delta} f(x)dx = \delta f(t) + \frac{\delta^2}{2} f'(t^*)$$
$$E(\hat{f}(t)) = f(t) + \frac{\delta}{2} f'(t^*)$$

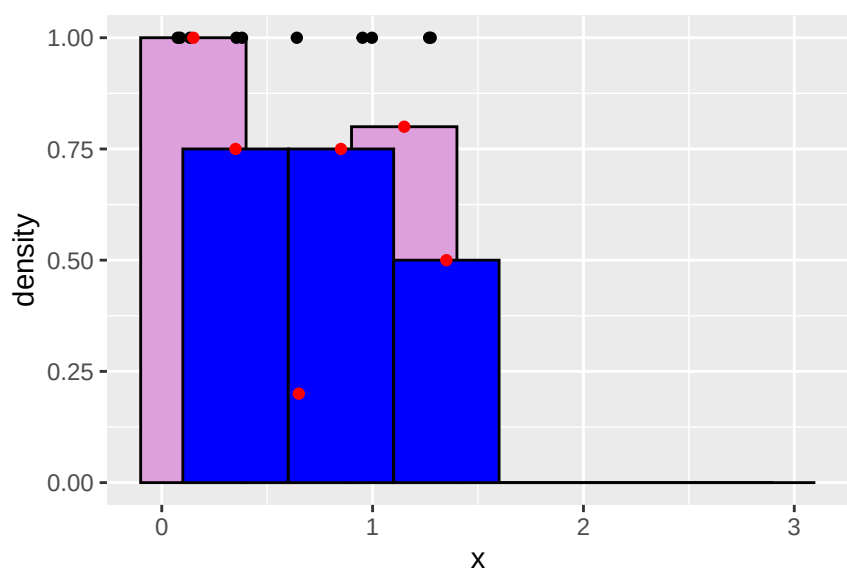
**So larger values of  $\delta$  give larger bias.**

The quality of an estimate (especially a biased estimate) can be measured using the expected (mean) squared error.

$MSE = Var + Bias^2$  We can find an optimal  $\delta$  (shall derive this later with kernel estimator) to minimize the MSE.

### 6.3 Shifted histogram

- We want alternative estimators.
- To remove the subjectivity in bin positioning, we take a fixed bin size and move it by very small amounts.
- As the bins move, the midpoints also shift.
- We define our new estimators at these midpoints.
- The estimator at the midpoint is the height of the histogram for that bin.



Now we superimpose with the actual density of  $\text{Gamma}(2,3)$ . Repeat the process with multiple data sets of size 10 from the same distribution.

### 6.4 Kernel density estimation

- We have solved the problem of bin locations by shifting the histograms.
- We have some idea about optimal bin widths through minimizing MSE.
- We still have the problems of non-smooth estimates for smooth parameter (density).
- This brings us to kernel density estimates that shift a smooth function instead of rectangular bins.
- In fact the shifting histogram is a special case of kernel density with a rectangular kernel.

- This is addressed by looking at the mean squared error.

$K(x)$  is a standard kernel function.

- Symmetric around 0.

$$K(x) = K(-x)$$

- It is a density.

$$\int K(x)dx = 1$$

- Hill shaped:  $K(x)$  is decreasing for  $x > 0$ .

- Mean 0 and Variance 1

$$\int x K(x)dx = 0 \quad \text{and} \quad \int x^2 K(x)dx = 1$$

The bandwidth  $h$  rescales the Kernel

$$K_h(x) = \frac{1}{h} K\left(\frac{x}{h}\right)$$

Still the same properties but variance =  $h^2$ .

#### **Different Kernels**

- Rectangular

$$\frac{1}{2} I(-1 < x < 1)$$

- Triangular

$$(1 - |x|) I(-1 < x < 1)$$

- Epanechnikov

$$\frac{3}{4} (1 - x^2) I(-1 < x < 1)$$

- Gaussian density

$$\frac{1}{\sqrt{2\pi}} \exp(-x^2)$$

#### **Kernel Density Estimator**

$$\hat{f}_h(t) = \frac{1}{n} \sum_{i=1}^n K_h(X_i - t)$$

This can be thought of the average of kernel functions at each  $t$  with center at the data points. (Remember the shifter histogram situation)

#### **Convolution**

$$f * g(x) = \int f(x - t)g(t)dt$$

If  $f_X(x)$  and  $g_Y(y)$  are the densities of independent random variables  $X$  and  $Y$  then the density of  $S = X + Y$  can be calculated by noting that if  $X = x$  and  $S = s$  then  $Y$  must be  $s - x$ . If we then integrate over all of the possible values of  $X$ ,

$$f_S(s) = \int f_X(x)g_Y(s - x)dx = f_X * g_Y(s)$$

The density of the sum of independent observations is the convolution of the marginal densities.

**Discrete Convolutions** Discrete convolutions replace the integral with a sum over a discrete set of values. If the function  $p(k)$  is a discrete probability function on the positive integers then

$$p * h(x) = \sum_{k=1}^{\infty} p(k)h(x - k)$$

This can be used to derive the probability mass function for a pair of independent discrete random variables.

**KDE as Convolution** The Kernel Density Estimator is

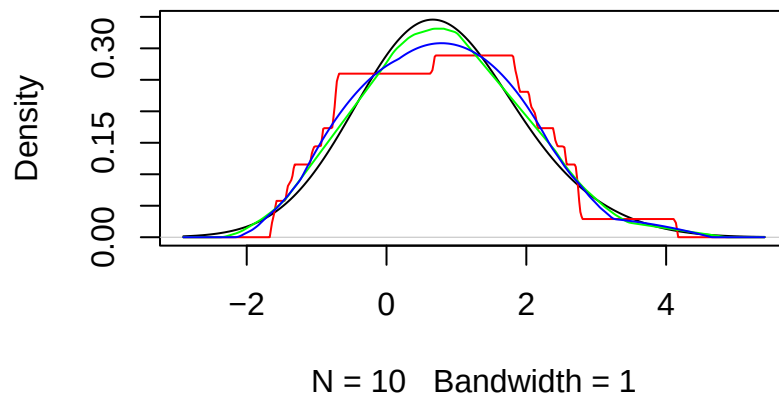
$$\hat{f}(t) = \frac{1}{n} \sum_{i=1}^n K_h(t - X_i)$$

Take the function  $\hat{p}(X_i)$  that puts discrete probability  $1/n$  on each observed point  $X_i$ . This is the empirical probability function that corresponds to the empirical CDF.

$$\hat{f}(t) = \sum_{i=1}^n K_h(t - X_i)\hat{p}(X_i) = K_h * \hat{p}(t)$$

**Smoothness** Convolution of a function with a smoother function retains the smoothness.

The smoothness property of the convolution implies that even if we have a discrete function  $\hat{p}$ , if  $K$  is a continuous kernel then the result will be continuous.



#### Observations

- Choosing a smooth kernel gives a smooth density estimate.
- The choice of kernel does not matter too much.
- Optimal choice of bandwidth remains subjective, as the minimize MSE results only give rates, not exact numbers.

## 7 Bandwidth Selection

Estimate the Density

$$\hat{f}_h(x) = \frac{1}{n} \sum_{i=1}^n K_h(x - X_i)$$

where the bandwidth  $h$  re-scales the kernel

$$K_h(d) = \frac{1}{h} K\left(\frac{d}{h}\right)$$

Bias as convolution

$$\text{Bias} = E(\hat{f}_h) - f = \int K_h(x - t)f(t)dt - f(x) = K_h * f - f$$

Notice that this doesn't depend on  $n$ .  
By Taylor expansion of  $f$  around  $x$ ,

$$f(t) = f(x) + (t - x)f'(x) + \frac{(t - x)^2}{2}f''(x) + o((t - x)^2)$$

$$\begin{aligned}
\int K_h(x-t)f(t)dt &= \int K_h(x-t)f(x)dt + \int K_h(x-t)(t-x)f'(x)dt \\
&\quad + \int K_h(x-t)\frac{(t-x)^2}{2}f''(x)dt + o(h^2) \\
&= f(x) \times 1 + f'(x) \times 0 + f''(x) \times \frac{h^2}{2} + o(h^2) \\
&= f(x) + f''(x)\frac{h^2}{2} + o(h^2)
\end{aligned}$$

Similarly we can show that

$$\text{Var}(\hat{f}_h(x)) = \frac{1}{nh}f(x) + o\left(\frac{1}{nh}\right)$$

Variance is inversely proportional to the number of observations in the bandwidth.

Find the average squared-error and integrate over the whole sample space. This gives the Mean Integrated Squared Error(MISE)

$$\begin{aligned}
\text{MISE}(\hat{f}) &= \int E\left(\hat{f}_h(x) - f(x)\right)^2 dx \\
&= \int \left( (E(\hat{f}_h(x)) - f(x))^2 + E\left(\hat{f}_h(x) - E(\hat{f})\right)^2 \right) dx \\
&= \int (\text{Bias}^2(x) + \text{Var}(x)) dx
\end{aligned}$$

MISE is minimized when  $h \propto n^{-1/5}$ . With this choice the optimal MISE is  $n^{-4/5}$ . This is a little worse than the parametric rate of  $1/n$ .

## 8 Squared Error Cross Validation

We want to estimate the integrated error

$$\int \left(\hat{f}_h(x) - f(x)\right)^2 dx$$

Expand the quadratic

$$\int \left(\hat{f}^2 - 2f\hat{f}_h + f^2\right) dx = \int \hat{f}_h^2(x)dx - 2 \int \hat{f}_h(x)f(x)dx + \int f^2(x)dx$$

The first term we can calculate from the data for each  $h$ .

The last term is the same for all  $h$ . We don't know what it is but we can safely ignore it.

The second term is the troublesome one, but notice that it is equal to  $-2E\hat{f}_h(X_1)$ . We can use an average to estimate this

$$\frac{1}{n} \sum_{i=1}^n \hat{f}_h(X_i)$$

But notice that this would be

$$\frac{1}{n} \sum_{i=1}^n \left( \frac{1}{n} \sum_{j=1}^n K_h(X_i - X_j) \right)$$

Fix  $i$  and take expectation of the inner quantity.

$E(K_h(X_i - X_j)) = E\hat{f}(X_i)$  if  $i \neq j$  and  $K_h(0)$  if  $i = j$

Thus expectation of the inner quantity is

$$\frac{n-1}{n} E\hat{f}(X_i)$$

We get a biased estimate.

Correct for this by removing this one observation from the estimation of the density

$$\hat{f}_{-i}(X_i) = \frac{1}{n-1} \sum_{j \neq i} K_h(X_i - X_j)$$

Thus

$$R(h) = \int \hat{f}_h^2(x) dx - \frac{2}{n} \sum_{i=1}^n \hat{f}_{-i}(X_i).$$

**Normal Reference Rule** If we have a normal density then the best bandwidth is  $1.06\sigma n^{-1/5}$ .

Here we have the exact constant of proportionality.

Of course this depends on unknown variance  $\sigma^2$ .

Plugging in the sample variance works reasonably well.

This bandwidth that will work for other smooth, unimodal densities, at least as a starting point.